

Data Availability Statement

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Notation Matters: Inconsistency of LLMs in Molecular Representations

Digital Discovery, Royal Society of Chemistry

Animesh Mishra | am847@snu.edu.in | ORCID: 0009-0009-1770-6329

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Public Repository

All experimental code, documentation, and article source are publicly available **without embargo** at:

<https://github.com/amethystani/notation-matters>

No authentication is required. The repository is licensed under the MIT License.

Available Artifacts

Experimental Scripts

`07_full_esol_phase1.py` Phase 1 — notation consistency evaluation across four molecular representations (SMILES, IUPAC, InChI, SELFIES) on 1072 molecules from the ESOL dataset.

`08_full_esol_phase2.py` Phase 2 — Ridge regression pooling of per-notation prediction heads; produces regression coefficients and cross-fold validation metrics.

`run_full_benchmark.sh` Batch pipeline that executes both phases sequentially with consistent random seeds.

Documentation and Article Source

- `README.md` — Installation instructions, usage guide, and reproducibility notes
- `main.tex` — Complete 6-page LaTeX article source (Communication format)
- `rsc.bib` — Bibliography file with all DOIs verified
- `DATA_AVAILABILITY_STATEMENT.md` — Machine-readable version of this statement

Source Dataset

Ground-truth aqueous solubility values ($\log S$) are from the publicly available ESOL dataset:

- Delaney, J. S. *J. Chem. Inf. Comput. Sci.* **2004**, *44*(3), 1000–1005.
- DOI: <https://doi.org/10.1021/ci034243x>

IUPAC names and InChI identifiers are retrieved at runtime via the PubChem REST API (<https://pubchem.ncbi.nlm.nih.gov/>).

Accessibility Summary

Public access	Yes — no login or request required
Embargo	None
Usage restrictions	None (MIT License)
Persistent URL	https://github.com/amethystani/notation-matters
Archived results	Generated at runtime by executing provided scripts
External dataset	ESOL via Delaney (2004), retrieved via PubChem REST API

Reproducibility Notes

- LLM inference uses Qwen3.6 via Ollama; minor numerical variation is possible across hardware configurations.
- All statistical tests (Fisher exact, Kruskal-Wallis, Mann-Whitney U, Wilcoxon signed-rank) use fixed random seeds for deterministic reproduction.
- 56 of 1128 ESOL molecules are excluded due to missing PubChem IUPAC entries; final dataset comprises 1072 molecules.
- 89.4% of SMILES strings yield valid SELFIES representations; the remainder are excluded from SELFIES analyses.

Contact

Queries regarding code execution or reproducibility will be addressed within five business days:

Animesh Mishra am847@snu.edu.in
School of Engineering, Shiv Nadar Institution of Eminence
<https://orcid.org/0009-0009-1770-6329>

I certify that all data and code referenced in the manuscript are publicly accessible at the repository above, without embargo or conditions of access, as of 28 May 2026.

Animesh Mishra
28 May 2026